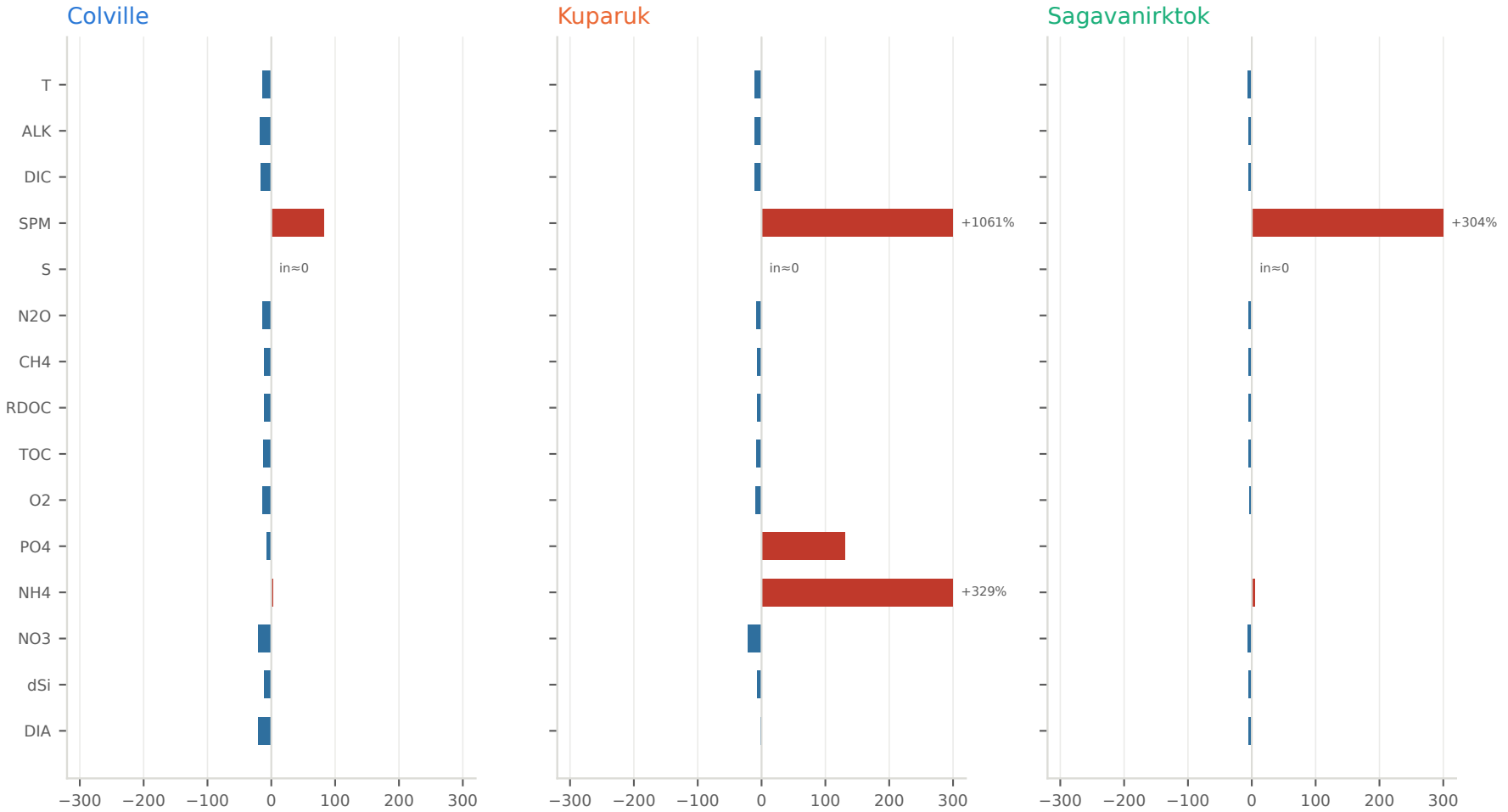
Estuarine filtering — % retained (−) / added (+), 1981–2023 mean



Percent change from input (upstream) to output (mouth) load, mean over 1981-2023, clipped at ±300% for display; 'in≈0' marks a near-zero input base (pct_change is not meaningful there -- see the CSV for the raw value). Orange = net source (added in the estuary), blue = net sink (removed). Upstream boundaries (except T, and TOC in the 44-yr record) are CONSTANT -- structure shown here is the model's own internal dynamics acting on a constant source, not a real time-varying river signal. See CLAUDE.md -> "Publication figure suite". This is an ADVECTIVE-flux-only estimate. For S (salinity) specifically, dispersive/tidal-pumping transport -- the dominant salt-intrusion mechanism on this coast (see CLAUDE.md -> "Width and dispersion") -- is NOT captured, so S's entry likely understates true salt exchange; the biogeochemical solutes are advection-dominated and less affected by this limitation.